

6. Evaluation Methodology

The supplied evaluation artifacts contain five questions, compared at $k = 1, 3, 5$, and 10 . Three questions have expected-source labels; two questions have no expected source and cannot contribute to a meaningful source-hit assessment. Hit rate is the fraction of labelled questions for which the expected source appears in the top- k retrieved sources. Retrieval latency measures the retrieval operation, not total LLM response time.

7. Results

7.1 Top-k retrieval evaluation

Top-k	Questions in summary	Hit rate	Mean retrieval latency
1	5	0.0%	16.22 ms
3	5	33.3%	15.89 ms
5	5	33.3%	16.51 ms
10	5	66.7%	16.42 ms

The recorded mean retrieval latency remains around 16 milliseconds across the tested top- k settings. Within this small evaluation, the source hit rate is highest at $k = 10$ (66.7%), compared with 33.3% at $k = 3$ and $k = 5$. The $k = 1$ run recorded 0%. This suggests that considering a larger candidate set increased the chance of including an expected source in this test set.

These metrics describe retrieval only; hit rate is not equivalent to factual answer accuracy. The test set is small and only three questions have expected-source labels.

7.2 Positive implementation observations

- The project configuration defines local Ollama embedding and generation models.
- Retrieval returns ranked passages with source filenames, chunk ranges, and similarity scores.
- When evidence for admission eligibility was absent from the retrieved context, the chatbot stated that it could not find the information rather than inventing official requirements.
- The recorded retrieval operation has low latency in the supplied evaluation, with mean values of approximately 15.9–16.5 ms.

7.3 Evidence-aware response behaviour

For the First Year B.Tech eligibility question, the chatbot explicitly stated that the available knowledge base did not contain the requested criteria. This is a useful cautious-answer behaviour, but it demonstrates abstention rather than successful retrieval of the admission answer. The retrieved passages in that example were not relevant to eligibility.

8. Interpretation and Inference

The experiment demonstrates an end-to-end RAG architecture with local embeddings, vector search, and local LLM generation. The evaluation illustrates the role of top-k selection: considering more candidates can improve the chance of finding the expected source, although irrelevant passages can still be included.

The strongest supported performance observation is retrieval speed: the mean retrieval latency is about 16 ms in the supplied test. Retrieval quality is preliminary because only three questions have expected-source labels.

Admission-related queries also show that the current index did not reliably surface the exact information requested. Corpus coverage, source freshness, metadata, and ranking are therefore important next areas for improvement.

The project is best described as a working prototype with low-latency retrieval and evidence-aware answer handling, rather than a fully validated admissions assistant. No unsupported accuracy percentage is claimed.

9. Conclusion

A domain-specific RAG chatbot for SPIT-related content was configured using a reported knowledge base of 101 PDFs and 100 website pages. It uses 450-word chunks with 60-word overlap, Ollama embeddings from nomic-embed-text, a FAISS vector index, and llama3.2 for response generation. The supplied evaluation records show retrieval latency of approximately 16 ms and a higher source hit rate when increasing top-k to 10 in this small test set.

The experiment also shows that successful execution is not the same as successful retrieval. Some admission queries did not surface relevant eligibility or document-requirement evidence. Acknowledging missing evidence is preferable to fabricating institutional requirements, but improved retrieval coverage and a larger labelled evaluation set are needed before claiming reliable domain performance.

10. Future Improvements

- Add and verify current official SPIT admission eligibility pages, required-document lists, and admission brochures.
- Preserve titles, dates, URLs, page numbers, and document types as searchable metadata.
- Down-rank stale announcements when users ask for current policy.
- Compare embedding models and chunking strategies.
- Expand the evaluation set with expected sources and reference answers.
- Measure answer faithfulness and relevance separately from retrieval latency.
- Check whether answer citations point to supporting passages.

11. Reproduction Commands

The following commands are included for reproducibility; they do not imply that you need to rerun ingestion for this report.

```
# Rebuild index after changing source content
uv run main.py ingest

# Ask a question
uv run main.py ask "What are the First Year B.Tech admission eligibility criteria at SPIT?"

# Inspect retrieved passages
uv run main.py retrieve "Documents required for admission to SPIT"

# Run top-k evaluation
uv run main.py evaluate
```

12. Viva Questions and Answers

Q1. What is RAG?

RAG combines retrieval from an external knowledge base with generation by a language model.

Q2. Why is RAG useful for institutional information?

The source documents can be updated without retraining the LLM, and responses can be tied to retrieved evidence.

Q3. What are embeddings?

Numerical vector representations of text that support semantic similarity search.

Q4. Why use chunk overlap?

It helps preserve context that crosses boundaries between adjacent chunks.

Q5. What does top-k mean?

The number of highest-ranked passages returned for a query.

Q6. What does hit rate measure?

Whether the expected source appears in the top-k retrieved sources for labelled questions; it does not measure answer correctness.

Q7. Why should the chatbot admit when evidence is missing?

It reduces the risk of presenting unsupported institutional claims as facts.

Q8. What is the main inference?

Retrieval was fast in the test, and a larger top-k improved expected-source coverage in the small sample; relevance still needs validation.

Q9. What is FAISS used for?

Efficient similarity search over vector embeddings.

Q10. What is a limitation of this evaluation?

Only five questions were tested and only three had expected-source labels, so broad accuracy cannot be established.

13. Repository and References

Source code: https://github.com/CodeCraftsmanRaj/GenAI/tree/main/Experiment_8_SPIT_RAG

Technical references

- Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS 2020.
- FAISS documentation, Facebook AI Similarity Search.
- Ollama documentation for local model and embedding APIs.
- Project config.yaml and supplied evaluation CSV/JSONL artifacts.

End of Report